VRS Observation Generation Algorithm

Derivation of the virtual reference station observation generation algorithm

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English edition

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1 Introduction

This document briefly derives the basic algorithm by which the observations of a Virtual Reference Station (VRS) are generated.

Assume that at epoch *t* the rover *u* and the reference stations *A*, *B* and *C* almost simultaneously observe the L1 C/A signals broadcast by *n* satellites, and let λ denote the L1 C/A wavelength. The pseudorange observations are

$$\begin{aligned}
\rho_u^i &= r_u^i + c\,dt_u - c\,dt^i + I_u^i + T_u^i + \epsilon_u^i \\
\rho_A^i &= r_A^i + c\,dt_A - c\,dt^i + I_A^i + T_A^i + \epsilon_A^i \\
\rho_B^i &= r_B^i + c\,dt_B - c\,dt^i + I_B^i + T_B^i + \epsilon_B^i \\
\rho_C^i &= r_C^i + c\,dt_C - c\,dt^i + I_C^i + T_C^i + \epsilon_C^i
\end{aligned} \tag{1}$$

where

$$i = 1, 2, \ldots, n \tag{2}$$

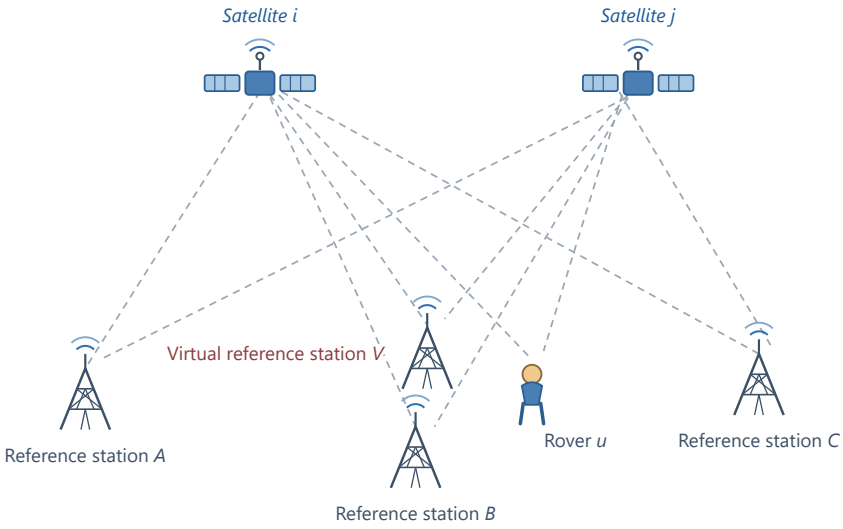


Figure 1 — Reference stations, virtual reference station, rover and satellites

2 Observation model assumptions

Taking satellite *j* as the reference satellite and *A* as the master reference station, assume that the double-difference ionospheric delays are described by spatial functions F_I evaluated at the station positions:

$$\begin{Bmatrix} I_u^i = F_I^i(r_u, r^i) \\ I_A^i = F_I^i(r_A, r^i) \\ I_B^i = F_I^i(r_B, r^i) \\ I_C^i = F_I^i(r_C, r^i) \end{Bmatrix} , \quad \begin{Bmatrix} I_u^j = F_I^j(r_u, r^j) \\ I_A^j = F_I^j(r_A, r^j) \\ I_B^j = F_I^j(r_B, r^j) \\ I_C^j = F_I^j(r_C, r^j) \end{Bmatrix} \tag{3}$$

and, in the same way, the tropospheric delays by functions F_T:

$$\begin{Bmatrix} T_u^i = F_T^i(r_u, r^i) \\ T_A^i = F_T^i(r_A, r^i) \\ T_B^i = F_T^i(r_B, r^i) \\ T_C^i = F_T^i(r_C, r^i) \end{Bmatrix} , \quad \begin{Bmatrix} T_u^j = F_T^j(r_u, r^j) \\ T_A^j = F_T^j(r_A, r^j) \\ T_B^j = F_T^j(r_B, r^j) \\ T_C^j = F_T^j(r_C, r^j) \end{Bmatrix} \tag{4}$$

The double-difference pseudorange observation equations of the rover and of the reference stations are then

$$\begin{aligned} \rho_{uA}^{ij} &= r_{uA}^{ij} + I_{uA}^{ij} + T_{uA}^{ij} + \epsilon_{uA}^{ij} \\ \rho_{BA}^{ij} &= r_{BA}^{ij} + I_{BA}^{ij} + T_{BA}^{ij} + \epsilon_{BA}^{ij} \\ \rho_{CA}^{ij} &= r_{CA}^{ij} + I_{CA}^{ij} + T_{CA}^{ij} + \epsilon_{CA}^{ij} \end{aligned} \tag{5}$$

3 Linearization about the master reference station

Given the known I_{BA}^{ij} and I_{CA}^{ij}, the rover double-difference ionosphere I_{uA}^{ij} is estimated as

$$\begin{aligned} I_{uA}^{ij} &= I_u^i - I_A^i - (I_u^j - I_A^j) = F_I^i(r_u, r^i) - F_I^i(r_A, r^i) - (F_I^j(r_u, r^j) - F_I^j(r_A, r^j)) \\ I_{BA}^{ij} &= I_B^i - I_A^i - (I_B^j - I_A^j) = F_I^i(r_B, r^i) - F_I^i(r_A, r^i) - (F_I^j(r_B, r^j) - F_I^j(r_A, r^j)) \\ I_{CA}^{ij} &= I_C^i - I_A^i - (I_C^j - I_A^j) = F_I^i(r_C, r^i) - F_I^i(r_A, r^i) - (F_I^j(r_C, r^j) - F_I^j(r_A, r^j)) \end{aligned} \tag{6}$$

Expanding $F_I^i(r_B, r^i)$ and $F_I^i(r_C, r^i)$ as Taylor series about r_A and neglecting the higher-order terms gives

$$\begin{aligned} F_I^i(r_B, r^i) &= F_I^i(r_A, r^i) + \frac{\partial F_I^i(r_A, r^i)}{\partial r_A^T} \cdot (r_B - r_A) \\ F_I^i(r_C, r^i) &= F_I^i(r_A, r^i) + \frac{\partial F_I^i(r_A, r^i)}{\partial r_A^T} \cdot (r_C - r_A) \end{aligned} \tag{7}$$

and, likewise, for $F_I^j(r_B, r^j)$ and $F_I^j(r_C, r^j)$:

$$\begin{aligned} F_I^j(r_B, r^j) &= F_I^j(r_A, r^j) + \frac{\partial F_I^j(r_A, r^j)}{\partial r_A^T} \cdot (r_B - r_A) \\ F_I^j(r_C, r^j) &= F_I^j(r_A, r^j) + \frac{\partial F_I^j(r_A, r^j)}{\partial r_A^T} \cdot (r_C - r_A) \end{aligned} \tag{8}$$

Substituting (6) and (7) into (5) yields

$$\begin{aligned} I_{BA}^{ij} &= \left(\frac{\partial F_I^i(r_A, r^i)}{\partial r_A^T} - \frac{\partial F_I^j(r_A, r^j)}{\partial r_A^T} \right) \cdot (r_B - r_A) \\ I_{CA}^{ij} &= \left(\frac{\partial F_I^i(r_A, r^i)}{\partial r_A^T} - \frac{\partial F_I^j(r_A, r^j)}{\partial r_A^T} \right) \cdot (r_C - r_A) \end{aligned} \tag{9}$$

that is,

$$\begin{bmatrix} I_{BA}^{ij} \\ I_{CA}^{ij} \end{bmatrix} = \begin{bmatrix} (r_B - r_A)^T \\ (r_C - r_A)^T \end{bmatrix} \underbrace{\left(\frac{\partial F_I^i(r_A, r^i)}{\partial r_A^T} - \frac{\partial F_I^j(r_A, r^j)}{\partial r_A^T} \right)^T}_{\chi_I^{ij}} \tag{10}$$

where the column χ_I^{ij} is the double-difference ionospheric gradient. Expanding $F_I^i(r_u, r^i)$ and $F_I^j(r_u, r^j)$ about r_A in the same way,

$$\begin{aligned} F_I^i(r_u, r^i) &= F_I^i(r_A, r^i) + \frac{\partial F_I^i(r_A, r^i)}{\partial r_A^T} \cdot (r_u - r_A) \\ F_I^j(r_u, r^j) &= F_I^j(r_A, r^j) + \frac{\partial F_I^j(r_A, r^j)}{\partial r_A^T} \cdot (r_u - r_A) \end{aligned} \tag{11}$$

gives

$$I_{uA}^{ij} = \left(\frac{\partial F_I^i(r_A, r^i)}{\partial r_A^T} - \frac{\partial F_I^j(r_A, r^j)}{\partial r_A^T} \right) \cdot (r_u - r_A) = (r_u - r_A)^T \cdot \chi_I^{ij} \tag{12}$$

that is, the rover double-difference ionosphere equals the projection of the gradient χ_I^{ij} onto the rover–master baseline vector.

4 Tropospheric delays

The same treatment applied to the tropospheric delay gives

$$\begin{bmatrix} T_{BA}^{ij} \\ T_{CA}^{ij} \end{bmatrix} = \begin{bmatrix} (r_B - r_A)^T \\ (r_C - r_A)^T \end{bmatrix} \underbrace{\left(\frac{\partial F_T^i(r_A, r^i)}{\partial r_A^T} - \frac{\partial F_T^j(r_A, r^j)}{\partial r_A^T} \right)^T}_{\chi_T^{ij}} \tag{13}$$

and

$$T_{uA}^{ij} = \left(\frac{\partial F_T^i(r_A, r^i)}{\partial r_A^T} - \frac{\partial F_T^j(r_A, r^j)}{\partial r_A^T} \right) \cdot (r_u - r_A) = (r_u - r_A)^T \cdot \chi_T^{ij} \tag{14}$$

5 Virtual pseudorange observations

Suppose that a virtual reference station *V* is placed near the rover, still using *A* as the master reference station. The double-difference delays at *V* follow directly from the gradients:

$$\begin{aligned} I_{VA}^{ij} &= (r_V - r_A)^T \cdot \chi_I^{ij} \\ T_{VA}^{ij} &= (r_V - r_A)^T \cdot \chi_T^{ij} \end{aligned} \tag{15}$$

and the VRS double-difference pseudorange observation equation is

$$\rho_{VA}^{ij} = r_{VA}^{ij} + I_{VA}^{ij} + T_{VA}^{ij} + \epsilon_{VA}^{ij} \tag{16}$$

Forming the *V–A* and *u–A* double differences of the pseudoranges gives

$$\begin{aligned} \rho_V^i - \rho_A^i - (\rho_V^j - \rho_A^j) &= r_V^i - r_A^i - (r_V^j - r_A^j) + I_{VA}^{ij} + T_{VA}^{ij} + \epsilon_{VA}^{ij} \\ \rho_u^i - \rho_A^i - (\rho_u^j - \rho_A^j) &= r_u^i - r_A^i - (r_u^j - r_A^j) + I_{uA}^{ij} + T_{uA}^{ij} + \epsilon_{uA}^{ij} \end{aligned} \tag{17}$$

Expanding r_u^i and r_u^j with r_V and re-grouping the delay terms,

$$\begin{aligned} \rho_u^i - \rho_A^i - (\rho_u^j - \rho_A^j) = r_u^i - (r_V^i + (r_A^i - r_V^i)) - (r_u^j - (r_V^j + (r_A^j - r_V^j))) \\ + (I_{uV}^{ij} + I_{uA}^{ij} - I_{uV}^{ij}) + (T_{uV}^{ij} + T_{uA}^{ij} - T_{uV}^{ij}) + \epsilon_{uV}^{ij} \end{aligned} \tag{18}$$

which can be written as

$$\underbrace{\rho_u^i - \rho_A^i + (r_V^i - r_A^i) + I_{VA}^{ij} + T_{VA}^{ij}}_{\rho_V^i} - \underbrace{(\rho_u^j - \rho_A^j + (r_V^j - r_A^j))}_{\rho_V^j} = r_u^i - r_V^i - (r_u^j - r_V^j) + I_{uV}^{ij} + T_{uV}^{ij} + \epsilon_{uV}^{ij} \tag{19}$$

From the above, the virtual pseudorange observations are

$$\begin{aligned}\rho_V^i &= \rho_A^i + (r_V^i - r_A^i) + I_{VA}^{ij} + T_{VA}^{ij} \\ \rho_V^j &= \rho_A^j + (r_V^j - r_A^j)\end{aligned} \tag{20}$$

Taking satellite 1 as the reference satellite of the VRS, its virtual observations are

$$\begin{bmatrix}\rho_V^1 \\ \rho_V^2 \\ \vdots \\ \rho_V^n\end{bmatrix} = \begin{bmatrix}\rho_A^1 + (r_V^1 - r_A^1) \\ \rho_A^2 + (r_V^2 - r_A^2) + I_{VA}^{21} + T_{VA}^{21} \\ \vdots \\ \rho_A^n + (r_V^n - r_A^n) + I_{VA}^{n1} + T_{VA}^{n1}\end{bmatrix} \tag{21}$$

If the rover uses satellite 2 as its own reference satellite, the double-difference pseudorange equation formed between the rover and the VRS is

$$\begin{aligned}&\rho_u^1 - (\rho_A^1 + (r_V^1 - r_A^1)) - (\rho_u^2 - (\rho_A^2 + (r_V^2 - r_A^2) + I_{VA}^{21} + T_{VA}^{21})) \\ &= r_u^1 - r_V^1 - (r_u^2 - r_V^2) + I_{uV}^{12} + T_{uV}^{12} + \epsilon_{uV}^{12}\end{aligned} \tag{22}$$

6 Equivalence with the rover–master double differences

Expanding (21) with the definitions (19) and (20),

$$\begin{aligned}\rho_u^1 - \rho_A^1 - (\rho_u^2 - \rho_A^2) &= r_u^1 - r_V^1 - (r_u^2 - r_V^2) + (r_V^1 - r_A^1) - (r_V^2 - r_A^2) + I_{uV}^{12} - I_{VA}^{21} + T_{uV}^{12} - T_{VA}^{21} + \epsilon_{uV}^{12} \\ &= r_u^1 - r_A^1 - (r_u^2 - r_A^2) + I_{uA}^{12} + T_{uA}^{12} + \epsilon_{uA}^{12}\end{aligned} \tag{23}$$

For any other satellite, e.g. satellite 3,

$$\begin{aligned}&\rho_u^3 - (\rho_A^3 + (r_V^3 - r_A^3) + I_{VA}^{31} + T_{VA}^{31}) - (\rho_u^2 - (\rho_A^2 + (r_V^2 - r_A^2) + I_{VA}^{21} + T_{VA}^{21})) \\ &= r_u^3 - r_V^3 - (r_u^2 - r_V^2) + I_{uV}^{32} + T_{uV}^{32} + \epsilon_{uV}^{32}\end{aligned} \tag{24}$$

$$\begin{aligned}&\rho_u^3 - \rho_A^3 - (\rho_u^2 - \rho_A^2) \\ &= r_u^3 - r_V^3 - (r_u^2 - r_V^2) + I_{uV}^{32} + T_{uV}^{32} + (r_V^3 - r_A^3) + I_{VA}^{31} + T_{VA}^{31} - (r_V^2 - r_A^2) - I_{VA}^{21} \\ &\quad - T_{VA}^{21} + \epsilon_{uV}^{32} \\ &= r_u^3 - r_A^3 - (r_u^2 - r_A^2) + I_{uV}^{32} + I_{VA}^{31} - I_{VA}^{21} + T_{uV}^{32} + T_{VA}^{31} - T_{VA}^{21} + \epsilon_{uV}^{32} \\ &= r_u^3 - r_A^3 - (r_u^2 - r_A^2) + I_{uA}^{32} + T_{uA}^{32}\end{aligned} \tag{25}$$

and the delay terms reduce as

$$\begin{aligned}I_{uV}^{32} + I_{VA}^{31} - I_{VA}^{21} &= I_u^3 - I_V^3 - (I_u^2 - I_V^2) + (I_V^3 - I_A^3 - (I_V^1 - I_A^1)) - (I_V^2 - I_A^2 - (I_V^1 - I_A^1)) \\ &= I_u^3 - I_A^3 - (I_u^2 - I_A^2) = I_{uA}^{32}\end{aligned} \tag{26}$$

By induction, the full set of rover–VRS double differences

$$\begin{bmatrix}\rho_{uV}^{12} = r_{uV}^{12} + I_{uV}^{12} + T_{uV}^{12} + \epsilon_{uV}^{12} \\ \rho_{uV}^{32} = r_{uV}^{32} + I_{uV}^{32} + T_{uV}^{32} + \epsilon_{uV}^{32} \\ \vdots \\ \rho_{uV}^{n2} = r_{uV}^{n2} + I_{uV}^{n2} + T_{uV}^{n2} + \epsilon_{uV}^{n2}\end{bmatrix} \equiv \begin{bmatrix}\rho_{uA}^{12} = r_{uA}^{12} + I_{uA}^{12} + T_{uA}^{12} + \epsilon_{uA}^{12} \\ \rho_{uA}^{32} = r_{uA}^{32} + I_{uA}^{32} + T_{uA}^{32} + \epsilon_{uA}^{32} \\ \vdots \\ \rho_{uA}^{n2} = r_{uA}^{n2} + I_{uA}^{n2} + T_{uA}^{n2} + \epsilon_{uA}^{n2}\end{bmatrix} \tag{27}$$

is equivalent to the set of rover–master double differences referred to satellite 2. Feeding the virtual observations (20) to a single-baseline RTK filter is therefore equivalent to processing the network solution directly at the rover.

Suppose now that n reference stations B_1, B_2, …, B_n operate around the master reference station A, each of them forming a baseline with A:

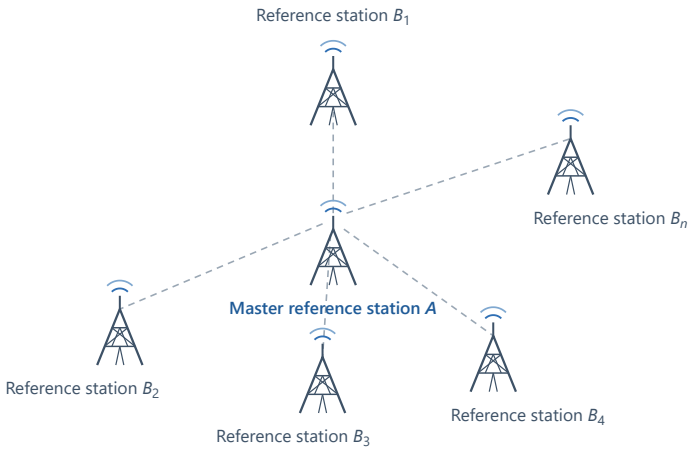


Figure 2 — Master reference station A and the surrounding reference stations $B_1 \ldots B_n$, forming the baselines AB_1, AB_2, …, AB_n

7 Network solution in the ENU frame

For the double-difference ionospheres of these baselines,

$$\begin{bmatrix} I^{ij}_{B_1A} \\ I^{ij}_{B_2A} \\ I^{ij}_{B_3A} \\ \vdots \\ I^{ij}_{B_nA} \end{bmatrix} = \begin{bmatrix} (r_{B_1} - r_A)^T \\ (r_{B_2} - r_A)^T \\ (r_{B_3} - r_A)^T \\ \vdots \\ (r_{B_n} - r_A)^T \end{bmatrix} \cdot \underbrace{\left(\frac{\partial F^i_I(r_A, r^i)}{\partial r^T_A} - \frac{\partial F^j_I(r_A, r^j)}{\partial r^T_A} \right)^T}_{\chi^{ij}_I} \tag{28}$$

To express χ_I^{ij} in the local ENU frame, let S be the rotation matrix from the ECEF frame to the topocentric coordinate system of the master station A. Inserting $S^T \cdot S = I$,

$$\begin{bmatrix} I^{ij}_{B_1A} \\ I^{ij}_{B_2A} \\ I^{ij}_{B_3A} \\ \vdots \\ I^{ij}_{B_nA} \end{bmatrix} = \begin{bmatrix} (r_{B_1} - r_A)^T \\ (r_{B_2} - r_A)^T \\ (r_{B_3} - r_A)^T \\ \vdots \\ (r_{B_n} - r_A)^T \end{bmatrix} \cdot S^T \cdot \underbrace{S \cdot \left(\frac{\partial F^i_I(r_A, r^i)}{\partial r^T_A} - \frac{\partial F^j_I(r_A, r^j)}{\partial r^T_A} \right)^T}_{\chi^{ij}_I} \tag{29}$$

Defining L_I^{ij}, the geometry matrix H and the ENU gradient $\chi_{I,ENU}^{ij}$ as

$$\underbrace{\begin{bmatrix} I_{B_1A}^{ij} \\ I_{B_2A}^{ij} \\ I_{B_3A}^{ij} \\ \vdots \\ I_{B_nA}^{ij} \end{bmatrix}}_{L_I^{ij}} = \underbrace{\begin{bmatrix} (S\cdot(r_{B_1}-r_A))^T \\ (S\cdot(r_{B_2}-r_A))^T \\ (S\cdot(r_{B_3}-r_A))^T \\ \vdots \\ (S\cdot(r_{B_n}-r_A))^T \end{bmatrix}}_{H} \cdot \underbrace{S\cdot\left(\frac{\partial F_I^i(r_A,r^i)}{\partial r_A^T} - \frac{\partial F_I^j(r_A,r^j)}{\partial r_A^T}\right)^T}_{\chi_{I,\mathrm{ENU}}^{ij}} \tag{30}$$

the observation model becomes

$$L_I^{ij} = H \cdot \chi_{I,\mathrm{ENU}}^{ij} \tag{31}$$

$\chi_{I,ENU}^{ij}$ contains the double-difference ionospheric gradients in the east, north and up directions. The least-squares estimate of (30) is

$$\chi_{I,\mathrm{ENU}}^{ij} = (H^T\cdot H)^{-1}\cdot H^T\cdot L_I^{ij} \tag{32}$$

The double-difference ionosphere of the virtual station *V* is finally computed as

$$I_{VA}^{ij} = (r_V - r_A)^T\cdot\chi_I^{ij} = \underbrace{(S\cdot(r_V-r_A))^T}_{r_{\mathrm{ENU}}^{VA}}\cdot(H^T\cdot H)^{-1}\cdot H^T\cdot L_I^{ij} = \left((H^T\cdot H)^{-1}\cdot H^T\cdot L_I^{ij}\right)^T\cdot r_{\mathrm{ENU}}^{VA} \tag{33}$$

and, in the same way, for the double-difference troposphere:

$$T_{VA}^{ij} = (r_V - r_A)^T\cdot\chi_T^{ij} = \left((H^T\cdot H)^{-1}\cdot H^T\cdot L_T^{ij}\right)^T\cdot r_{\mathrm{ENU}}^{VA} \tag{34}$$

8 Virtual carrier-phase observations

The derivation for the carrier-phase observations proceeds along the same lines. The carrier-phase observation equation is

$$\begin{aligned} \varphi_u^i &= r_u^i + c\,dt_u - c\,dt^i - I_u^i + T_u^i + N_u^i + \varepsilon_u^i \\ \varphi_A^i &= r_A^i + c\,dt_A - c\,dt^i - I_A^i + T_A^i + N_A^i + \varepsilon_A^i \\ \varphi_B^i &= r_B^i + c\,dt_B - c\,dt^i - I_B^i + T_B^i + N_B^i + \varepsilon_B^i \\ \varphi_C^i &= r_C^i + c\,dt_C - c\,dt^i - I_C^i + T_C^i + N_C^i + \varepsilon_C^i \end{aligned} \tag{35}$$

Taking satellite *j* as the reference satellite and *A* as the master reference station, the double-difference carrier-phase observation equations are

$$\begin{aligned} \varphi_{uA}^{ij} &= r_{uA}^{ij} - I_{uA}^{ij} + T_{uA}^{ij} + N_{uA}^{ij} + \epsilon_{uA}^{ij} \\ \varphi_{BA}^{ij} &= r_{BA}^{ij} - I_{BA}^{ij} + T_{BA}^{ij} + N_{BA}^{ij} + \epsilon_{BA}^{ij} \\ \varphi_{CA}^{ij} &= r_{CA}^{ij} - I_{CA}^{ij} + T_{CA}^{ij} + N_{CA}^{ij} + \epsilon_{CA}^{ij} \end{aligned} \tag{36}$$

As with the virtual pseudoranges, the virtual carrier-phase observations are

$$\begin{aligned} \varphi_V^i &= \varphi_A^i + (r_V^i - r_A^i) - I_{VA}^{ij} + T_{VA}^{ij} \\ \varphi_V^j &= \varphi_A^j + (r_V^j - r_A^j) \end{aligned} \tag{37}$$

and, with satellite 1 as the VRS reference satellite,

$$\begin{bmatrix}\varphi_V^1\\ \varphi_V^2\\ \vdots\\ \varphi_V^n\end{bmatrix} = \begin{bmatrix}\varphi_A^1 + (r_V^1 - r_A^1)\\ \varphi_A^2 + (r_V^2 - r_A^2) - I_{VA}^{21} + T_{VA}^{21}\\ \vdots\\ \varphi_A^n + (r_V^n - r_A^n) - I_{VA}^{n1} + T_{VA}^{n1}\end{bmatrix} \tag{38}$$

9 Second-order extension

If required, $F_I^i(r_u, r^i)$ and $F_I^j(r_u, r^j)$ can be expanded as Taylor series to second order:

$$\begin{aligned} F_I^i(r_u, r^i) &= F_I^i(r_A, r^i) + \frac{\partial F_I^i(r_A, r^i)}{\partial r_A^T}\cdot(r_u - r_A) + (r_u - r_A)^T \cdot \frac{\partial^2 F_I^i(r_A, r^i)}{\partial r_A \partial r_A}\cdot(r_u - r_A)\\ F_I^j(r_u, r^j) &= F_I^j(r_A, r^j) + \frac{\partial F_I^j(r_A, r^j)}{\partial r_A^T}\cdot(r_u - r_A) + (r_u - r_A)^T \cdot \frac{\partial^2 F_I^j(r_A, r^j)}{\partial r_A \partial r_A}\cdot(r_u - r_A)\end{aligned} \tag{39}$$

The rover double-difference ionosphere then becomes

$$\begin{aligned} I_{uA}^{ij} &= \left(\frac{\partial F_I^i(r_A, r^i)}{\partial r_A^T} - \frac{\partial F_I^j(r_A, r^j)}{\partial r_A^T}\right)\cdot(r_u - r_A) + (r_u - r_A)^T \cdot \left(\frac{\partial^2 F_I^i(r_A, r^i)}{\partial r_A \partial r_A} - \frac{\partial^2 F_I^j(r_A, r^j)}{\partial r_A \partial r_A}\right)\cdot(r_u - r_A)\\ &= (r_u - r_A)^T \cdot \left(\frac{\partial F_I^i(r_A, r^i)}{\partial r_A^T} - \frac{\partial F_I^j(r_A, r^j)}{\partial r_A^T}\right)^T + (r_u - r_A)^T \cdot \left(\frac{\partial^2 F_I^i(r_A, r^i)}{\partial r_A \partial r_A} - \frac{\partial^2 F_I^j(r_A, r^j)}{\partial r_A \partial r_A}\right)\cdot(r_u - r_A)\end{aligned} \tag{40}$$

$$= (r_u - r_A)^T \cdot \underbrace{\left(\frac{\partial F_I^i(r_A, r^i)}{\partial r_A^T} - \frac{\partial F_I^j(r_A, r^j)}{\partial r_A^T}\right)^T}_{\chi_I^{ij}} + (r_u - r_A)^T \cdot \underbrace{\left(\frac{\partial^2 F_I^i(r_A, r^i)}{\partial r_A \partial r_A} - \frac{\partial^2 F_I^j(r_A, r^j)}{\partial r_A \partial r_A}\right)}_{\psi_I^{ij}}\cdot(r_u - r_A) \tag{41}$$

Neglecting the correlation of the gradient between the horizontal directions,

$$\psi_I^{ij} = \frac{\partial^2 F_I^i(r_A, r^i)}{\partial r_A \partial r_A} - \frac{\partial^2 F_I^j(r_A, r^j)}{\partial r_A \partial r_A} \approx \begin{bmatrix}a & 0 & 0\\ 0 & b & 0\\ 0 & 0 & c\end{bmatrix} \tag{42}$$

so that

$$I_{uA}^{ij} = (r_u - r_A)^T \cdot \chi_I^{ij} + \underbrace{\left[(r_u - r_A)_x^2 \;\; (r_u - r_A)_y^2 \;\; (r_u - r_A)_z^2\right]}_{\tau_u^A}\cdot\begin{bmatrix}a\\ b\\ c\end{bmatrix} \tag{43}$$

The master reference station *A* again forms the baselines AB_1, AB_2, ..., AB_n with the surrounding stations. For their double-difference ionospheres,

$$\underbrace{\begin{bmatrix}I_{B_1A}^{ij}\\ I_{B_2A}^{ij}\\ I_{B_3A}^{ij}\\ \vdots\\ I_{B_nA}^{ij}\end{bmatrix}}_{L_I^{ij}} = \underbrace{\begin{bmatrix}(S\cdot(r_{B_1} - r_A))^T & \tau_{B_1}^A\\ (S\cdot(r_{B_2} - r_A))^T & \tau_{B_2}^A\\ (S\cdot(r_{B_3} - r_A))^T & \tau_{B_3}^A\\ \vdots & \\ (S\cdot(r_{B_n} - r_A))^T & \tau_{B_n}^A\end{bmatrix}}_{\Psi}\cdot\begin{bmatrix}\chi_{I,\mathrm{ENU}}^{ij}\\ a\\ b\\ c\end{bmatrix} \tag{44}$$

i.e.

$$L_I^{ij} = \Phi \cdot \begin{bmatrix} \chi_{I,\,\mathrm{ENU}}^{ij} \\ a \\ b \\ c \end{bmatrix} \tag{45}$$

The least-squares estimate of (44) is

$$\begin{bmatrix} \chi_{I,\,\mathrm{ENU}}^{ij} \\ a \\ b \\ c \end{bmatrix} = (H^T \cdot H)^{-1} \cdot H^T \cdot L_I^{ij} \tag{46}$$

The double-difference ionosphere of the virtual station *V* is then

$$\begin{aligned} I_{VA}^{ij} &= (r_V - r_A)^T \cdot \chi_I^{ij} + (r_u - r_A)^T \cdot (r_u - r_A) \cdot [a \;\; b \;\; c]^T \\ &= (S \cdot (r_V - r_A))^T \cdot \chi_{I,\,\mathrm{ENU}}^{ij} + \tau_{VA} \cdot [a \;\; b \;\; c]^T \\ &= \left[(S \cdot (r_V - r_A))^T \;\; \tau_{VA}\right] \cdot (\Phi^T \cdot \Phi)^{-1} \cdot \Phi^T \cdot L_I^{ij} \end{aligned} \tag{47}$$

The troposphere is handled in exactly the same way.